

Exponential and Logarithmic Function (147)

Exponential and logarithmic functions are inverse of each other.

Exponential Function

For a fixed value of the base a the expression a^x varies with the value of x .

e.g. $(0.5)^x$, 1^x , 2^x , 3^x and so on.

The function a^x (as different values are chosen for a) are called exponential functions.

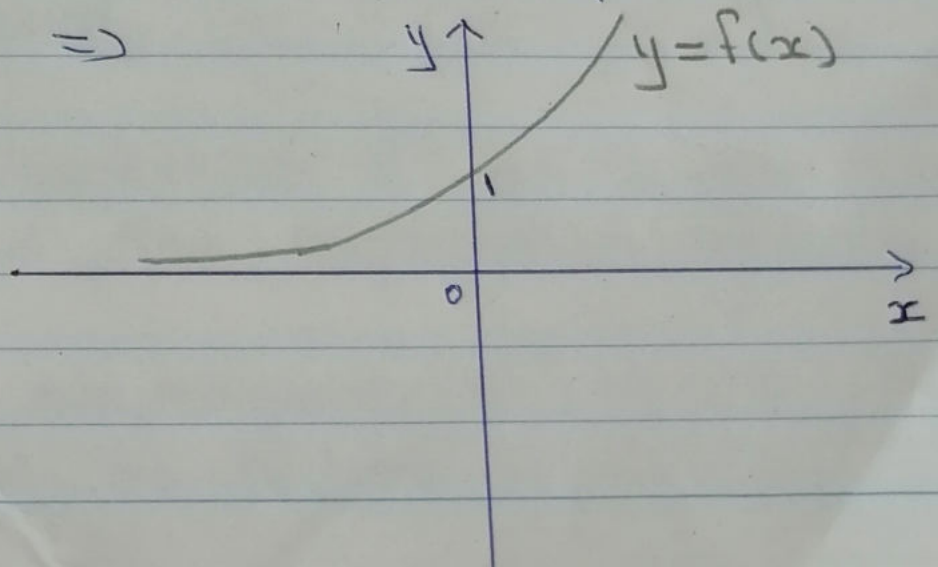
The most and widely used exponential function has the particular base

$$e = 2.7182818 \dots$$

Recall that e is an irrational function
 $\Rightarrow$ It cannot be expressed as a fraction.

The approximate value of e is stored in most calculators.

Let $f(x) = e^x$ be the exponential function $\Rightarrow$



(48)

The domain of an exponential function is $(-\infty, \infty)$ and the range is $(0, +\infty)$

Again an exponential function is continuous, increasing and differentiable.

Exponential functions appear in various areas of Mathematics and engineering.

For example, the shape of a hanging chain or rope, under the effect of gravity is well described by a combination of the exponential curves e^{kx} , e^{-kx} .

The function e^{-x^2} plays a major role in statistics; it being fundamental in the important normal distribution which describes many naturally occurring phenomena.

The exponential function e^{-kx} appears again in statistics, in the Poisson distribution which (amongst other things) is used to predict the number of events which occur randomly in a given time interval.

NB: The expression e^x as $x \rightarrow \infty$ is called the exponential growth

The expression e^{-x} as $x \rightarrow \infty$ is called the exponential decay.

Notice that e^x as $x \rightarrow \infty = \infty$

and e^{-x} as $x \rightarrow \infty = 0$.

$$\text{i.e. } e^{-x} = \frac{1}{e^x} = \frac{1}{e^\infty} = 0$$

NB: e denotes the unique positive real number such that $\ln e = 1$.

The natural Logarithmic function

The natural exponential function is the inverse of the natural logarithm.

The natural logarithm is to base e , denoted by $\ln$.

$$\Rightarrow \ln x = \ln_e x$$

Since this function is the inverse of the exponential function

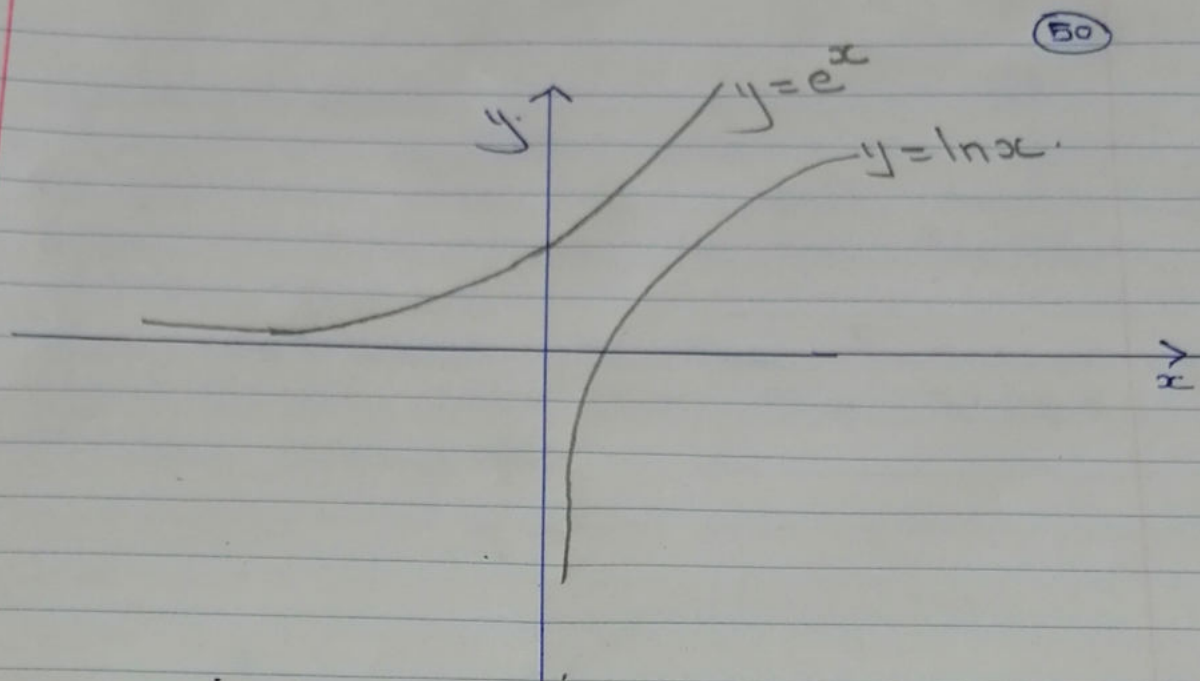
$$\Rightarrow y = \exp x \text{ if and only if } x = \ln y.$$

$$\text{i.e. } y = e^x \text{ iff } x = \ln y.$$

$$\text{Notice that } \ln y = x$$

$$\Rightarrow \ln e^x = x$$

$$\Rightarrow e^x = y.$$



The domain of the natural logarithm is $(0, +\infty)$ and the range is $(-\infty, \infty)$.

Again, like the exponential function, logarithm function is continuous and differentiable on its domain.

Theorem

- (i) $\ln e^x = x$ for every x .
- (ii) $e^{\ln x} = x$ for every $x > 0$.

Properties of $\ln x$.

If $a > 0$ and $b > 0$, then

(a) $\ln(ab) = \ln a + \ln b$.

(b) $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$.

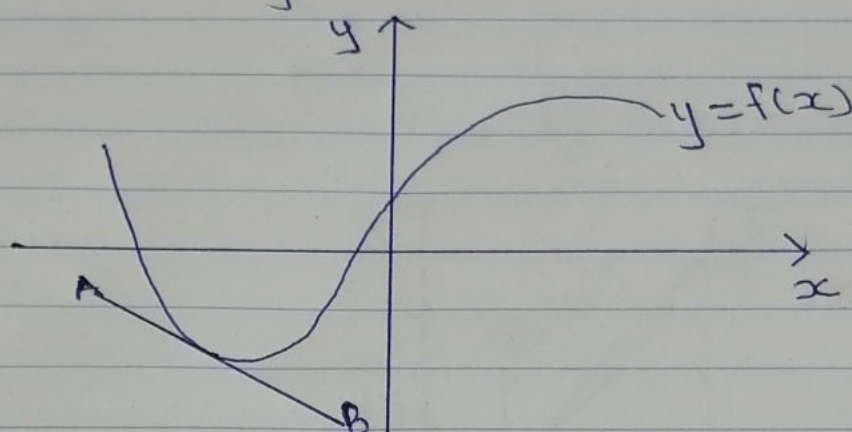
(c) $\ln a^r = r \ln a$.

Differentiation

(51)

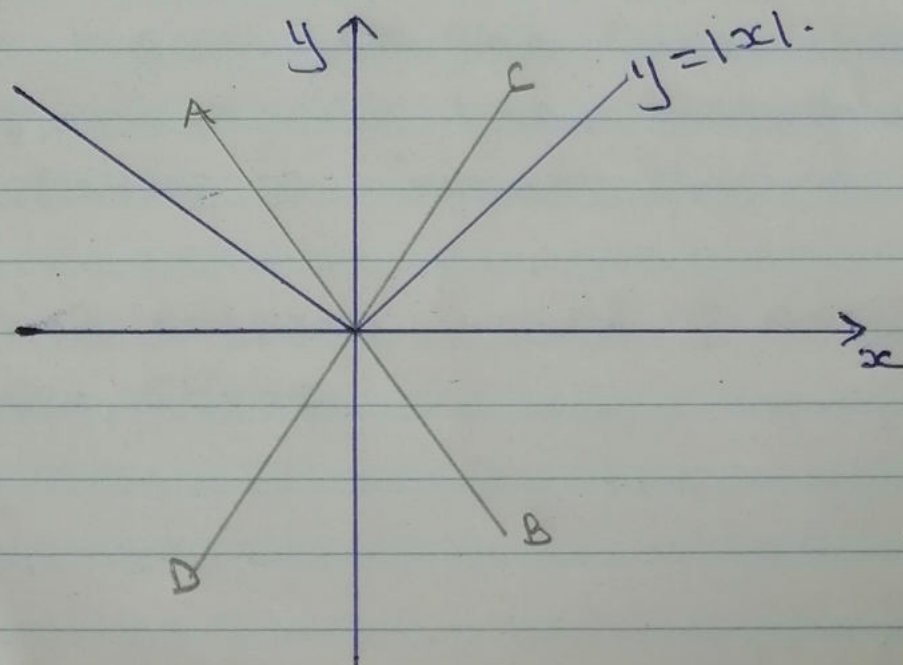
A function is differentiable at a point if we can draw a unique tangent line to the graph of the function, at that point.

The derivative of the function at that point will be the slope of the tangent.



For certain functions, even though they are continuous, this may not be possible.

For example, consider the function $y=f(x)=|x|$.

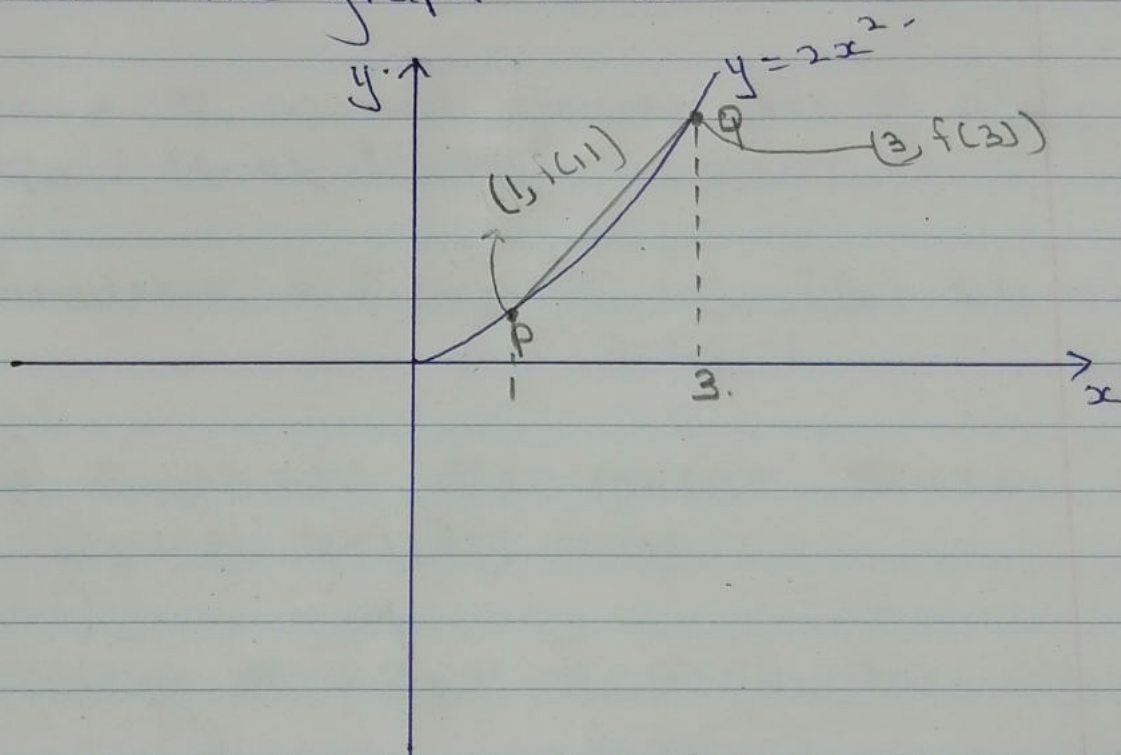


(52)

$\Rightarrow$ At $x=0$, the function $y=|x|$ is continuous, but it is not differentiable at $x=0$

NB: Continuity does not imply differentiability, but differentiability implies continuity.

Consider the graph of $f(x) = 2x^2$.



If Q is a point on the curve different from P , then the line through P and Q is called a Secant line.

This line rotates around P as Q moves along the curve.

Considering the above graph.

(53)

$$\text{Gradient (slope) of } PQ = \frac{\Delta y}{\Delta x} = \frac{f(3) - f(1)}{3 - 1} = \frac{18 - 2}{2} = 8$$

Now, Suppose Q moves closer to P , to $Q_1 = (2.5, f(2.5))$

$$\Rightarrow \text{Slope of } PQ_1 = \frac{f(2.5) - f(1)}{2.5 - 1} = 7.$$

Suppose Q_1 moves closer to P at $Q_2 = (1.01, f(1.01))$

$$\Rightarrow \text{Gradient of } PQ_2 = \frac{f(1.01) - f(1)}{1.01 - 1} = 4.02.$$

Lastly, Suppose Q_2 moves closer to P at $Q_3 = (1.001, f(1.001))$

$$\Rightarrow \text{Gradient of } PQ_3 = \frac{f(1.001) - f(1)}{1.001 - 1} = 4.002$$

Thus as the point Q approaches closer and closer to the point P , the gradient of the chords nears to the value 4.

$$\text{i.e. } \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = 4.$$

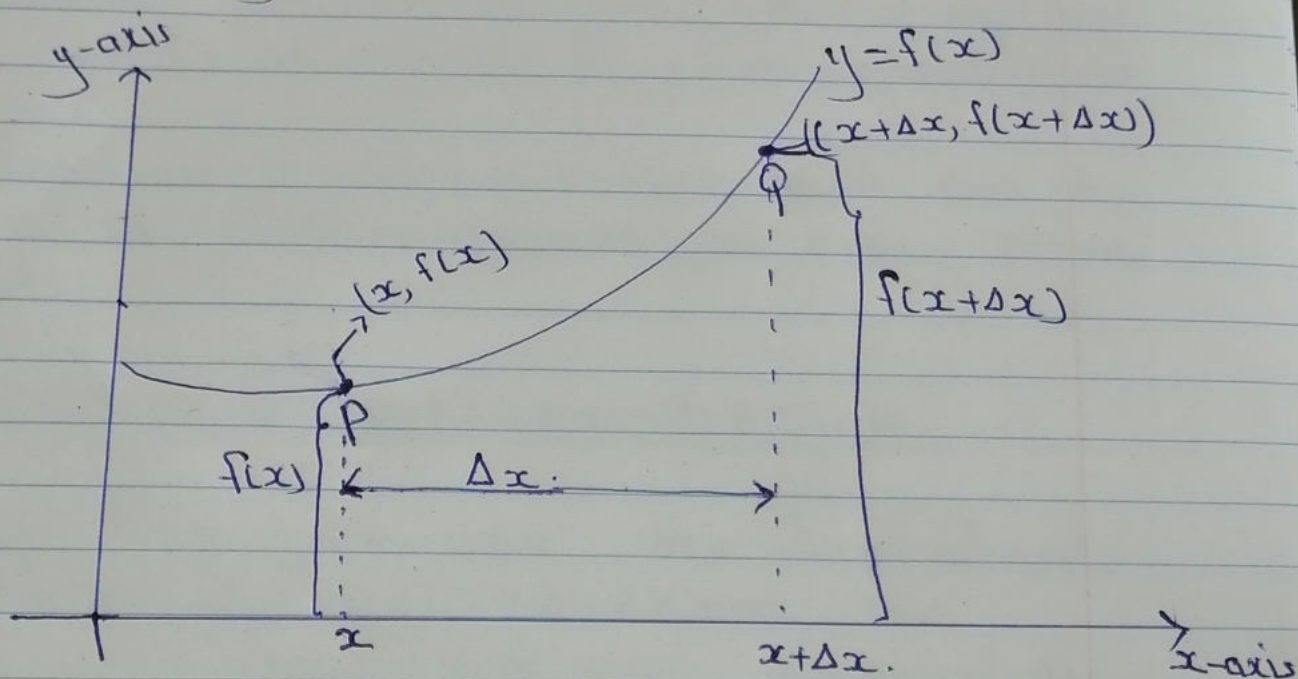
(54)

At the point P , the chord becomes the tangent to the curve.

Thus, the limiting value of Δy is the gradient of the tangent at P .

Differentiation from 1st principle.

Let P and Q be two points very close together on a curve.



$$\Rightarrow \text{Slope of } PQ = \frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

As $\Delta x \rightarrow 0$, the closer the gradient of the chord approaches the gradient of the tangent P .

$$\Rightarrow \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = \frac{d(f(x))}{dx} = f'(x)$$

at all points x for which the limit exists. (55)

If $f'(x)$ exists, then we say that f is differentiable at x .

NB: The domain of f' is the subset of $\text{dom}(f)$ where f' is defined.

$\text{Dom}(f')$ may be smaller than $\text{dom}(f)$ because it contains only those points in $\text{dom}(f)$ at which f is differentiable.

A function is differentiable on a set A if it is differentiable at every point $x \in A$.

NOTATION OF DIFFERENTIATION.

If $y = f(x)$, we use the dependent variable y to represent the function, and we can denote the derivative of the function with respect to x in any of the following ways.

$$\Delta_x y = y' = \frac{dy}{dx} = \frac{d}{dx} f(x) = f'(x) = D_x f(x) = Df(x).$$

The Symbol $\frac{d}{dx}$ is a differential operator

i.e the derivative with respect to x .

Examples

(56)

1. Find the derivative of the given function from the 1st principle.

(a) $f(x) = x^2$.

Solution

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x}$$

Note $f(x) = x^2$
 $f(x+\Delta x) = (x+\Delta x)^2$

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{(x+\Delta x)^2 - x^2}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{x^2 + 2x\Delta x + \Delta x^2 - x^2}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} 2x + \Delta x = 2x + 0 = 2x.$$

(b) $f(x) = 3x^3$.

Solution

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x+\Delta x) - f(x)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{3(x+\Delta x)^3 - 3x^3}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{3(x^3 + 3x^2\Delta x + 3x\Delta x^2 + \Delta x^3) - 3x^3}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{9x^2\Delta x + 9x\Delta x^2 + 3\Delta x^3}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} 9x^2 + 9x\Delta x + 3\Delta x^2 = 9x^2 + 0 + 0 = 9x^2.$$

2. If $f(x) = \frac{4x^3 - 8x^2 + 6x}{2x}$, find the

co-ordinates of the point at which the gradient is

(i) zero (ii) 4.

Solution

$$f(x) = \frac{4x^3 - 8x^2 + 6x}{2x} = 2x^2 - 4x + 3.$$

$$\Rightarrow f'(x) = 4x - 4.$$

$$(i) f'(x) = 4x - 4 = 0$$

$$\Rightarrow 4x = 4 \Rightarrow x = 1.$$

$$\text{When } x = 1, y = 2(1)^2 - 4(1) + 3 = 1.$$

$$\Rightarrow \text{at } (1, 1), \text{ gradient} = 0.$$

$$(ii) f'(x) = 4x - 4 = 4$$

$$4x = 8 \Rightarrow x = 2.$$

$$\text{When } x = 2, y = 2(2^2) - 4(2) + 3 = 3.$$

At the point $(2, 3)$, the gradient = 4.

Exercise:

1. Find the derivative of the following functions from 1st principle

$$(a) f(x) = \sqrt{x}$$

$$\text{Ans} = \frac{1}{2} x^{-1/2} = \frac{1}{2\sqrt{x}}.$$

$$(b) f(x) = \frac{1}{2x}$$

$$\text{Ans} = \frac{-1}{(2x)^2}$$

$$(c) f(x) = \frac{1}{5x+3}$$

$$\text{Ans} = \frac{-5}{(5x+3)^2}$$